
COSC 2671 — Social Media and Network Analytics

Assignment 2 — Group Project Report

Do Polarized Emotional Communities Form Around AI Job Replacement Discussions on YouTube, and Who Are the Influential Users Shaping These Communities?

Chantelle Isabelle Dmonte

Student ID: s4186963

Utkarsh Nautiyal

Student ID: s4190295

Post Graduate Group 39

1. Introduction

The rapid diffusion of artificial intelligence into the global workforce has generated widespread public debate. From op-eds in mainstream media to grass-roots video commentary, citizens, economists, and policymakers are actively negotiating the meaning and implications of AI-driven automation. YouTube, as the world's largest video-sharing and comment platform, provides a uniquely rich window into this ongoing conversation: its comment threads capture reactions from diverse global audiences who engage directly with video content on topics ranging from robot factory deployments to Senate floor speeches by legislators such as Bernie Sanders.

This study asks a focused empirical question: Do polarized emotional communities form around AI job replacement discussions on YouTube, and who are the influential users shaping these communities? The question is motivated by growing academic and policy interest in how algorithmic platforms either amplify or moderate affective polarisation around technological disruption. Understanding whether communities with systematically different sentiment orientations cluster in YouTube's reply graph — and whether identifiable "influencer" nodes bridge or entrench these divides — has practical implications for how platform designers, educators, and policymakers approach public communication about AI.

To address the research question we collected 10,100 YouTube comments across 77 videos and 873 Bluesky posts, built directed weighted reply networks for both platforms, applied VADER sentiment analysis and Latent Dirichlet Allocation (LDA) topic modelling to the text, performed Louvain community detection on the networks, and tested statistical hypotheses about sentiment polarisation across detected communities. The cross-platform comparison with Bluesky enables us to examine whether the polarisation pattern is platform-specific or a more general feature of online AI discourse.

1.1 Research Question and Success Criteria

The primary research question is: Do polarized emotional communities form around AI job replacement discussions on YouTube, and who are the influential users shaping these communities?

We defined four measurable success criteria at the outset:

- Detect at least two distinct communities with statistically different mean sentiment scores (polarisation = YES or NO answer, tested with Kruskal-Wallis and Mann-Whitney U tests).
- Identify and classify the top influential users per platform using a role taxonomy (Authority, Hub, Bridge, Peripheral) derived from multiple centrality measures.
- Demonstrate that topic modelling reveals distinct discursive frames (e.g., economic fear, techno-optimism, policy debate) and that these frames distribute differently across communities.
- Compare polarisation patterns between YouTube and Bluesky to assess platform-level differences in affective community structure.

2. Data Collection and Description

2.1 YouTube Data

YouTube comment data was collected using the YouTube Data API v3 via a custom Python client (data_collection/youtube_collector.py). Data collection targeted videos addressing AI and automation job displacement, retrieved through five keyword queries: "AI replacing jobs", "robots replace workers", "automation unemployment", "AI taking over jobs", and "artificial intelligence job loss". For each matching video, all available top-level comment threads and their replies were fetched using the commentThreads.list endpoint, paginating through all result pages.

Data was collected in May 2026. The final dataset comprises 10,100 comments across 77 YouTube videos. Key fields used in analysis include: comment_id, author_id (YouTube channel ID), text, like_count, published_at, is_reply (Boolean), and parent_author_id (for constructing reply edges). The 77 videos span a wide range of

channels — from individual creators and news organisations (CNN, CNBC, ABC News, BBC) to political figures (Senator Bernie Sanders) and viral comedy shorts — ensuring diversity in the discourse captured.

The dataset supports both network analysis (reply edges between `author_id` nodes) and NLP analysis (raw comment text). A representative sample (`sample_data/youtube_comments.csv`, 10,100 rows) is included in the submission. No API keys are included in submitted files.

2.2 Bluesky Data

Bluesky posts were collected using the AT Protocol public search API through a custom client (`data_collection/bluesky_collector.py`). The same five keyword queries were issued against the `app.bsky.feed.searchPosts` endpoint. In total, 873 posts were collected, with fields including `author_handle`, `author_did`, `text`, `created_at`, `like_count`, `reply_count`, `repost_count`, `reply_to_uri`, and `reply_to_handle`. Reply-to relationships are used to construct the Bluesky reply network.

Bluesky is a decentralised microblogging platform with significantly smaller but technically-engaged user base relative to YouTube. Its inclusion enables a cross-platform comparison and tests whether the polarisation patterns observed on YouTube are platform-specific or reflective of broader discourse dynamics around AI employment.

2.3 Data Limitations

Several limitations are acknowledged. First, YouTube's public API does not expose full historical archives; comments may have been removed by creators or flagged by YouTube prior to collection. Second, the Bluesky dataset is substantially smaller (873 posts vs. 10,100 YouTube comments), which limits the statistical power of Bluesky-specific network analyses. Third, both datasets reflect searches conducted at a specific point in time; sentiment patterns may shift as AI news cycles evolve. Fourth, YouTube comment authorship is identified by channel ID (not display name), so some content creators who also comment cannot be distinguished from ordinary commenters. Fifth, the VADER lexicon, while well-suited to short social media text, may misclassify technical or ironic content.

Attribute	YouTube	Bluesky
Collection method	YouTube Data API v3	AT Protocol public API
Collection date	May 2026	May 2026
Total records	10,100 comments	873 posts
Videos / queries	77 videos	5 keyword queries
Key text field	comment text	post text
Network edge type	reply-to (author → parent author)	reply-to (handle → handle)
NLP support	Full comment text	Full post text

Table 1. Summary of data sources and collection attributes.

3. Methodology

3.1 NLP Pipeline

3.1.1 Text Preprocessing

All text was preprocessed using the custom `nlp_analysis/preprocessor.py` module, which applies the following pipeline in sequence: (1) lowercasing; (2) URL removal using regular expressions; (3) removal of non-ASCII characters and excessive punctuation; (4) tokenisation using NLTK's `word_tokenize`; (5) stopword removal (NLTK English stopwords list, augmented with domain-specific terms such as "video" and "watch"); (6)

lemmatisation using NLTK's WordNetLemmatizer. The cleaned text is stored in a `clean_text` column for downstream analysis.

3.1.2 Sentiment Analysis (VADER)

Sentiment scoring was performed using VADER (Valence Aware Dictionary and sEntiment Reasoner), implemented via the `vaderSentiment` Python package. VADER was chosen because it is specifically designed for short, informal social media text, handles negations, intensifiers, and emoji-adjacent punctuation, and requires no training data. VADER outputs four scores per text: `pos`, `neg`, `neu`, and a compound score in the range $[-1, +1]$. A comment is labelled Positive if `compound` ≥ 0.05 , Negative if `compound` ≤ -0.05 , and Neutral otherwise. These thresholds follow the standard VADER guidelines (Hutto & Gilbert, 2014).

3.1.3 Topic Modelling (LDA)

Latent Dirichlet Allocation was used to identify latent thematic clusters in the text corpus. The `CountVectorizer` was configured with `max_df=0.95`, `min_df=3`, `max_features=5000`, and `ngram_range=(1,2)` to capture both unigrams and important bigrams. LDA was run with `n_components=5` (five topics), `max_iter=15`, and `learning_method="online"`. The number of topics was chosen as five to balance interpretability with coverage of the discourse.

To validate the LDA model beyond simple topic coherence, a train/test perplexity split (80%/20%) was performed. Perplexity — measuring how well the model generalises to unseen text — yielded 3,732.61 for YouTube and 5,771.32 for Bluesky (lower = better fit). Each document is assigned its dominant topic (the topic with highest probability), stored in the `yt_dominant_topic` and `bsky_dominant_topic` columns.

The five YouTube topics and their most representative terms are:

- Topic 0 — Economic & Systems Framing: company | way | system | lot | day | future | money | mean | thing | using
- Topic 1 — Robot Anxiety / Technology Skepticism: robot | need | thing | something | machine | right | actually | always | video | long
- Topic 2 — General Optimism / Acceptance: good | much | new | everyone | point | use | every | life | thanks | sure
- Topic 3 — Economic Inequality / Policy (Wealth & Tax): money | pay | need | never | want | back | rich | tax | worker | cost
- Topic 4 — Human vs. Machine / Replacement Debate: human | year | time | take | replace | robot | yes | world | real | want

The five Bluesky topics are:

- Topic 0 — White-Collar Displacement: taking | replacing | collar | white | human | layoff | company | white collar | value | good
- Topic 1 — Billionaire Accountability (Bezos/Sanders frame): million | billion | worker | company | bezos | robot | jeff bezos | jeff | replace | bernie
- Topic 2 — Tech Infrastructure / Tax Policy: replacing | worker | human | need | data | replace | system | robot | buy | tax
- Topic 3 — Amazon/Musk Robotics Deployment: robot | worker | replace | replace worker | worker robot | robot replace | tax | amazon | musk | human
- Topic 4 — Automation & Unemployment: automation | unemployment | taking | tech | year | loss | replacing | massive | take | many

3.2 Network Construction

For both platforms, directed weighted reply networks were constructed using `NetworkX`. Each node represents a unique user (identified by YouTube channel ID or Bluesky handle/DID). A directed edge from node A to node B is added when user A replied directly to a comment or post by user B; the edge weight counts the

number of such reply interactions. Self-loops are removed, and users with missing or "unknown" identifiers are excluded.

Design decisions are justified as follows. The network is directed because replies have an unambiguous sender and receiver — the direction encodes the flow of responsive attention. The network is weighted because repeated replies between two users indicate a stronger or more persistent interaction, and weighted edges allow degree and centrality measures to reflect interaction intensity rather than merely binary connection. A threshold of minimum edge weight ≥ 1 is applied (all reply relationships, regardless of frequency, are included), because the overall network is already sparse and further pruning would fragment it.

Property	YouTube Network	Bluesky Network
Nodes (users)	9,808	51
Edges (reply pairs)	8,830	36
Network type	Directed, weighted	Directed, weighted
Density	0.000092	0.014118
Avg. clustering coefficient	0.0022	0.0000
Avg. shortest path length	14.54	1.85
Diameter	51	2
Reciprocity	0.0002	0.0000
Weakly connected components	1,257	15

Table 2. Network summary statistics for YouTube and Bluesky reply graphs.

3.3 Network Analysis (COSC 2671 PG Requirement)

Five centrality measures were computed for all nodes in both networks, chosen to capture complementary dimensions of influence:

- In-degree centrality: the proportion of all possible incoming edges that point to a given node. High in-degree indicates a node is replied to by many others — a signal of conversational salience or authority.
- Betweenness centrality: the fraction of all shortest paths in the network that pass through a given node. High betweenness identifies "bridge" users who connect otherwise disconnected communities and through whom information must flow between clusters.
- PageRank: a recursive importance measure that weights incoming links by the PageRank of their sources. Being replied to by already-influential users yields a higher PageRank than being replied to by peripheral users, capturing prestige rather than raw popularity.
- Eigenvector centrality: similar to PageRank but based on the adjacency matrix eigenvector. It rewards connections to high-centrality neighbours.
- HITS (Hyperlink-Induced Topic Search): separates hub score (pointing to high-authority content) from authority score (being pointed to by high-hub nodes). HITS is particularly appropriate here because reply networks have natural hub-authority structure — some users generate many replies to prominent commenters (hubs) while others attract replies from hub users (authorities).

User roles are classified using a composite taxonomy (`influence_roles.py`): Authorities exhibit high in-degree and authority score; Hubs show high hub score and out-degree; Bridges have high betweenness and moderate degree; Peripheral users score low on all measures. This taxonomy enables nuanced interpretation of community influence beyond a simple centrality ranking.

3.4 Community Detection

The Louvain algorithm (python-louvain package) was applied to the undirected projection of each reply graph to detect communities. Louvain maximises modularity Q , which measures the density of edges within communities relative to a random null model. The algorithm is particularly suitable here because it is parameter-free (no need to specify the number of communities), scales efficiently to networks of thousands of nodes, and has well-established use in social network analysis literature (Blondel et al., 2008).

For the YouTube network, Louvain detected 1,293 communities (including many small isolate-clusters of 1-2 users). For Bluesky, 15 communities were detected. Community membership was merged with the analysed text data to enable per-community sentiment profiling.

3.5 Polarisation Testing

To answer the primary research question statistically, we applied two non-parametric tests that do not assume normally distributed sentiment scores. The Kruskal-Wallis H-test was first applied across all detected communities (null hypothesis: all communities have the same distribution of VADER compound scores). A significant result ($p < 0.05$) indicates that at least one community differs from the others. To further confirm polarisation between the most extreme communities, Mann-Whitney U was applied between the most positive community (mean sentiment = 0.750, community 350) and the most negative (mean sentiment = -0.365, community 1205).

4. Results and Analysis

4.1 Sentiment Analysis Results

Across the 10,100 YouTube comments, VADER classified 4,524 (44.8%) as Positive, 2,621 (26.0%) as Negative, and 2,955 (29.3%) as Neutral. The mean compound score was positive (approximately +0.12, $\sigma = 0.461$), suggesting a slight overall optimistic lean in YouTube commentary about AI job replacement — though the substantial proportion of negative comments indicates significant concern. This pattern likely reflects the mixture of content on YouTube: viral robot-showcase videos and comedy clips tend to attract lighter engagement, while more sober policy-focused videos (such as the Sanders Senate floor speech) attract more critical comments.

The Bluesky corpus of 873 posts showed a markedly different profile: 332 (38.0%) Positive, 434 (49.7%) Negative, and 107 (12.3%) Neutral. The mean compound score was negative ($\sigma = 0.513$). Bluesky's more left-leaning, policy-oriented user base and its decentralised structure apparently select for more critical engagement with AI displacement narratives, particularly around corporate accountability (topics 1 and 3 above).

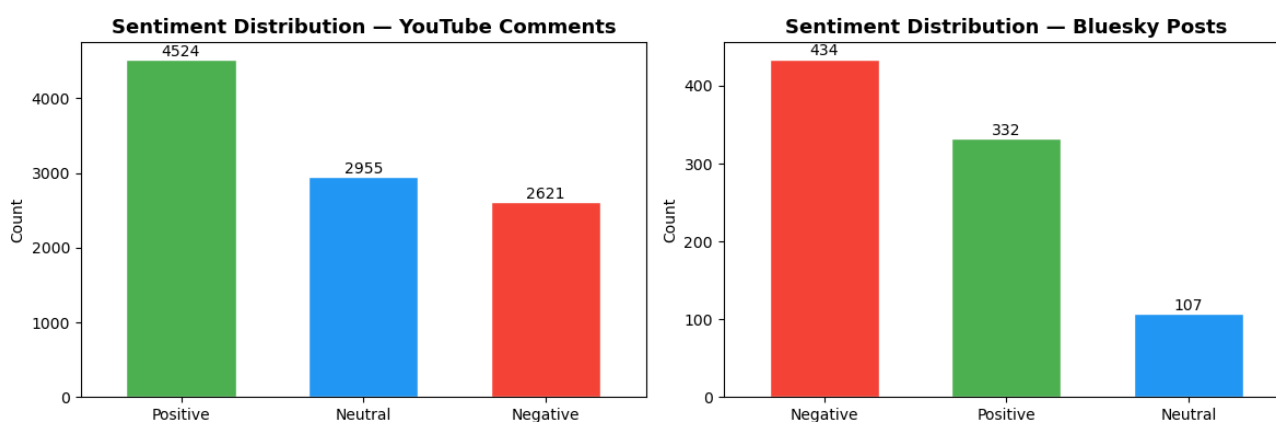


Figure 1. Sentiment distribution (Positive / Neutral / Negative) across YouTube (left) and Bluesky (right). YouTube shows a positive lean while Bluesky skews negative.

4.2 Topic Modelling Results

LDA topic modelling identified five latent topics on each platform. On YouTube, the topics range from abstract economic concern (Topic 0: system, future, company) to concrete robot-replacement anxiety (Topic 1), techno-optimism or casual acceptance (Topic 2), economic inequality and wealth redistribution debate (Topic 3: rich, tax, worker), and the philosophical human-vs-machine framing (Topic 4: human, replace, robot). Topic 3 is notable for its policy orientation — the presence of "tax" and "rich" alongside "worker" suggests awareness of redistribution debates stimulated by politicians like Sanders. Topics 1 and 4 show the highest overlap in language (both involve robot and replace) but diverge in framing: Topic 1 focuses on technological skepticism, Topic 4 on existential comparison.

On Bluesky, the topics are more tightly clustered around specific corporate actors and political economy: Bezos and Amazon appear prominently in Topics 1 and 3, Senator Bernie Sanders is present, and white-collar displacement features explicitly in Topic 0. This reflects Bluesky's user base, which tends to engage with political and economic critiques of tech corporations rather than the general viral AI content prevalent on YouTube.

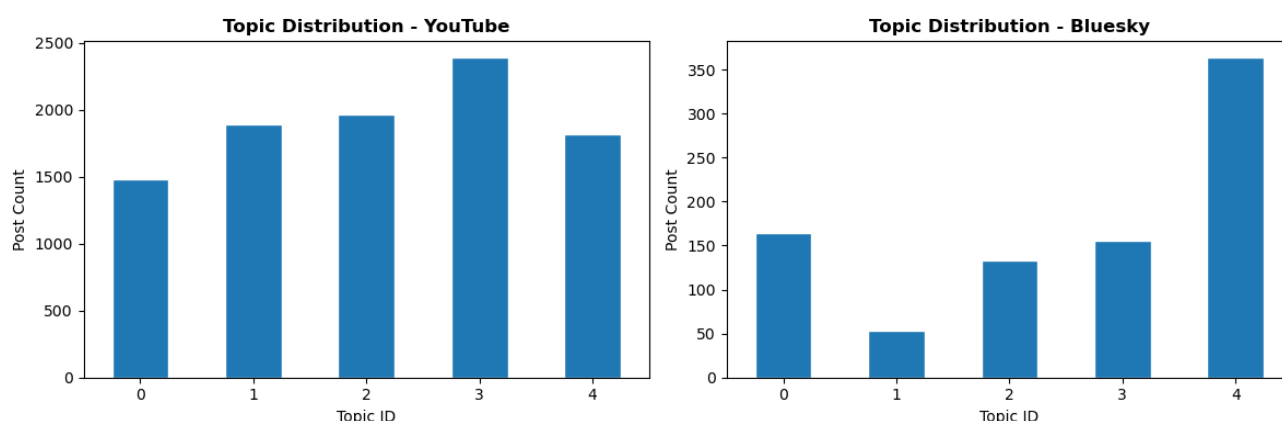


Figure 2. LDA topic distribution across YouTube (left) and Bluesky (right). Topic proportions indicate the dominant discursive frames in each dataset.

4.3 Network Analysis Results

4.3.1 Network Structure

The YouTube reply network is large and sparse: 9,808 nodes (users) and 8,830 directed reply edges, with density 0.000092. The low average clustering coefficient (0.0022) and high diameter (51) indicate that the network is structured as many small local clusters (discussion threads) weakly connected through a small number of bridge users. The average in-degree and out-degree are equal at 0.900, reflecting the predominantly dyadic (one-reply) interaction structure of YouTube comment sections. The network has 1,257 weakly connected components, consistent with the fact that most comment threads are isolated within individual videos.

The Bluesky network is much smaller (51 nodes, 36 edges, density 0.014) and less cohesive: the diameter is only 2 (direct sender-to-receiver paths), and there are 15 weakly connected components across 51 users. The near-zero clustering suggests users are replying without triggering further reply chains — a reflection of Bluesky's smaller user count and less developed thread culture in this topic area.

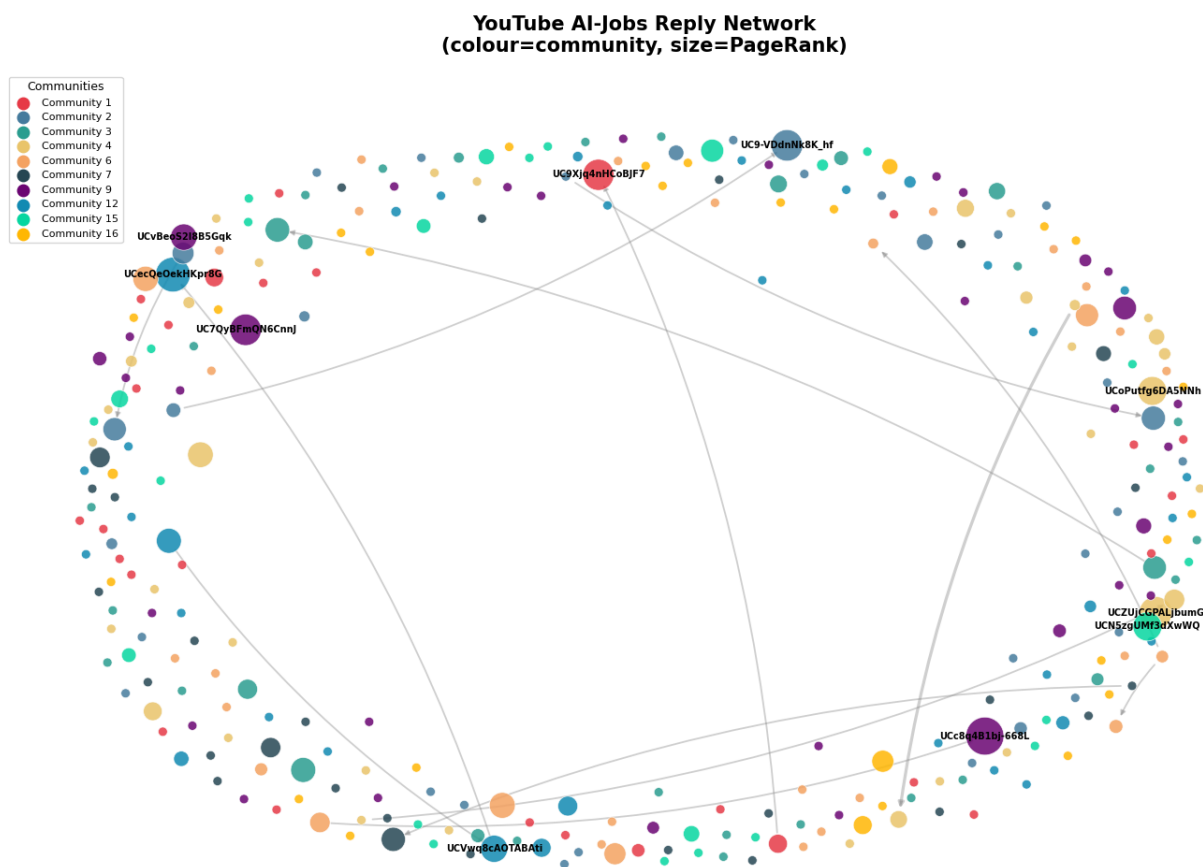


Figure 4. YouTube reply network visualisation. Node colour indicates community membership; node size reflects in-degree centrality. The sarse, fragmented structure reflects the many isolated comment threads across 77 videos.

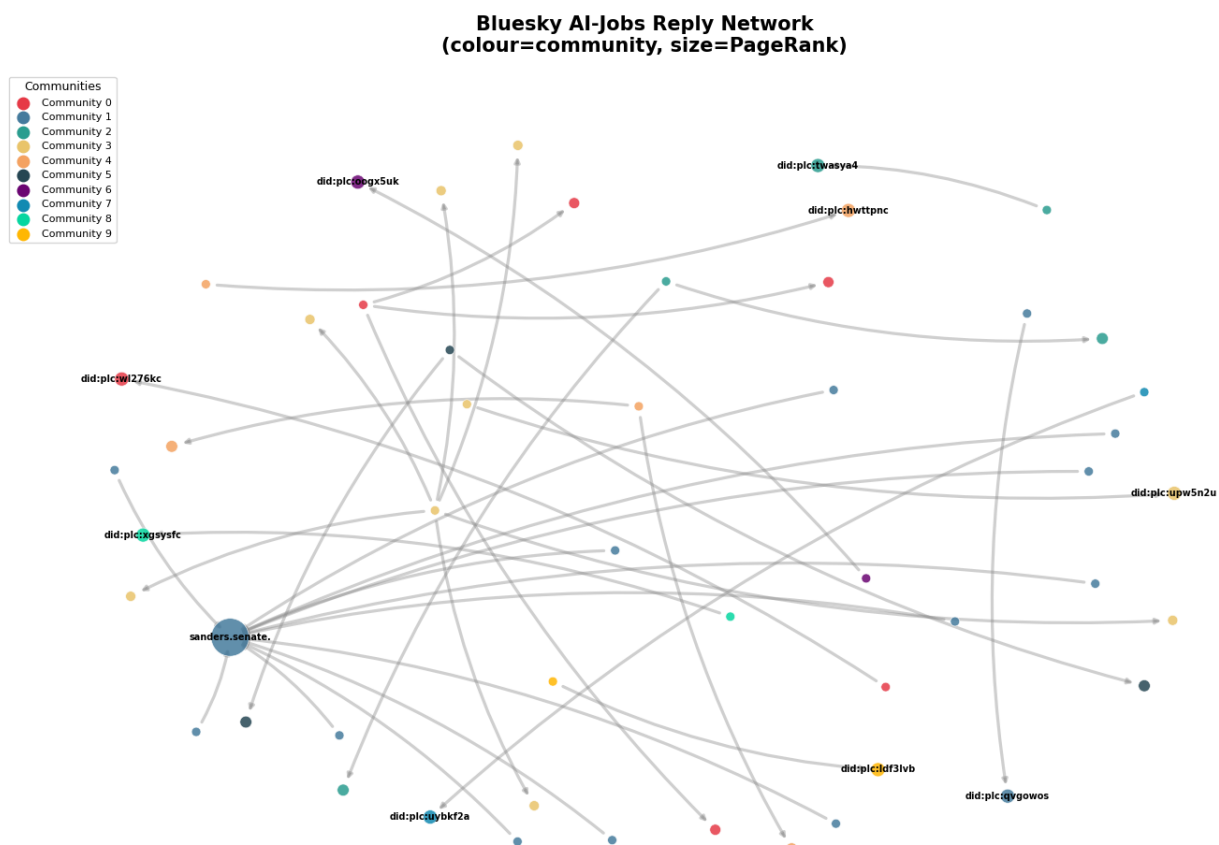


Figure 5. Bluesky reply network visualisation. The small network (51 nodes) shows limited thread connectivity but clear community separation. *sanders.senate.gov* dominates as the sole Authority node.

4.3.2 Centrality Analysis

The centrality scatter plot (Figure 3) visualises the relationship between PageRank (x-axis) and betweenness centrality (y-axis), with node size proportional to in-degree. Most nodes cluster near the origin, confirming the network's highly skewed influence distribution. A handful of nodes stand out with elevated PageRank, corresponding to channel IDs that receive a disproportionate share of reply traffic — these are primarily large channels whose videos attract substantial comment-reply interaction.

Betweenness centrality is extremely low across the YouTube network (maximum observed betweenness ≈ 0.0045), because the network's fragmented structure means there are very few cross-component shortest paths for any node to lie on. This is an expected structural property of YouTube comment networks, where the reply graph is essentially a forest of shallow trees (video comment sections) rather than a densely interconnected social network.

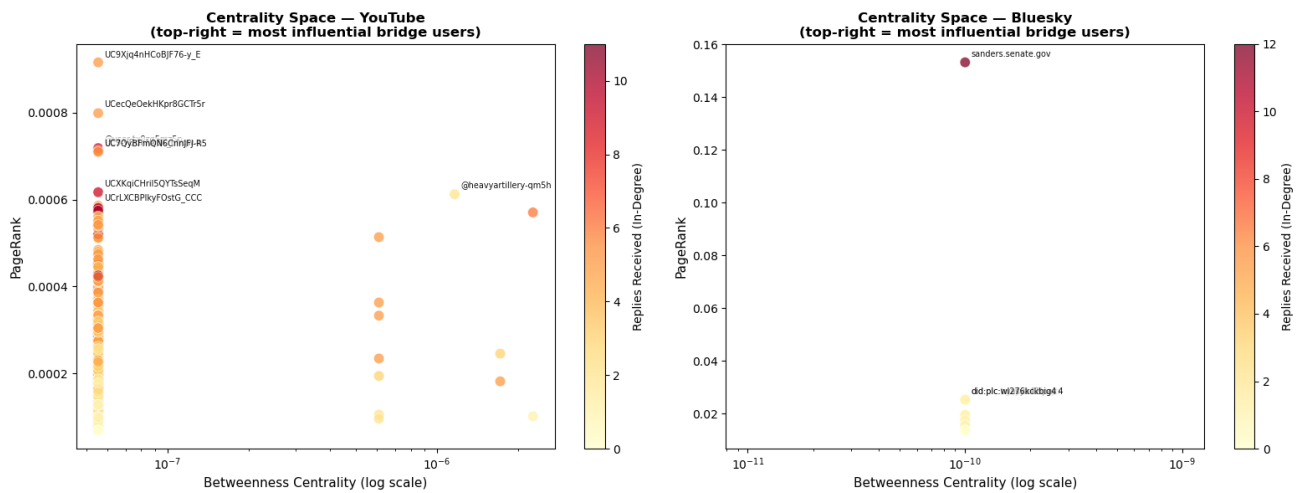


Figure 3. PageRank vs. betweenness centrality scatter for YouTube (left) and Bluesky (right). Node size = normalised in-degree. Most nodes are peripheral; a small number of Authority nodes receive disproportionate reply attention.

4.3.3 Influence Roles

Role classification using the composite taxonomy identifies four categories of users. The results for the top influential users are summarised in Table 3 and visualised in Figure 7.

Rank	Platform	User ID / Handle	Role	PageRank	In-degree
1	YouTube	UCc8q4B1bj-668LMHyNXnTxQ	Authority	0.000818	0.909
2	YouTube	UCoPutfg6DA5NNhmG5eybLhQ	Authority	0.000600	0.909
3	YouTube	UCN5zgUMf3dXwWQ117tf4WiQ	Authority	0.000600	0.909
4	YouTube	UCvBeoS2l8B5GqkSr-wXR_Qg	Authority	0.000551	0.909
5	YouTube	UC9-VDdnNk8K_hfrqw-nwAlw	Authority	0.000656	0.636
6	YouTube	UCVwq8cAOTABAti58Eg21aAQ	Hub	0.000564	0.182
1	Bluesky	sanders.senate.gov	Authority	0.153217	1.000
2	Bluesky	goodstephen.bsky.social	Hub	0.013680	0.000
3	Bluesky	seanofthesea.bsky.social	Hub	0.013680	0.000
4	Bluesky	russelltaylorphd.bsky.social	Hub	0.013680	0.000

Table 3. Top influential users on YouTube and Bluesky by role classification.

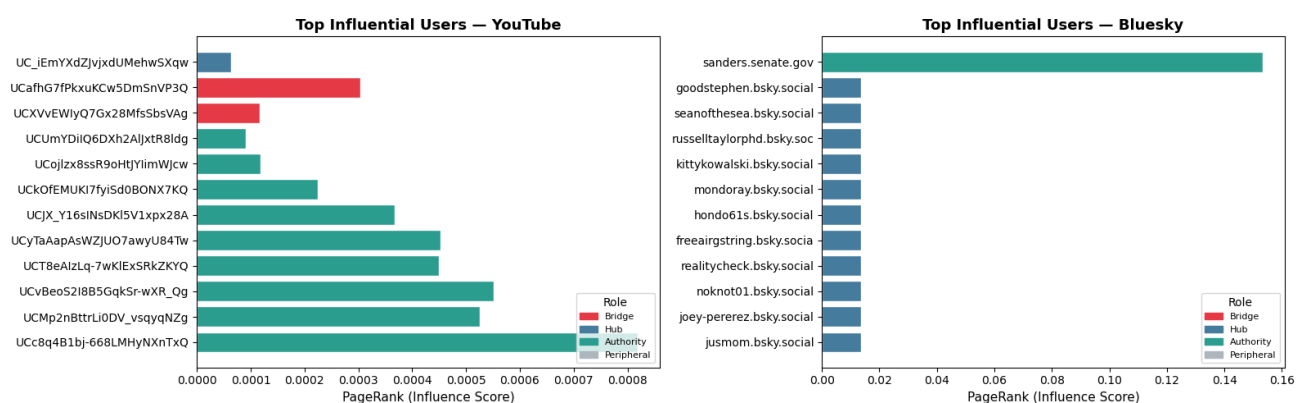


Figure 7. Top influential users visualisation. Bar heights represent composite influence score. Bluesky's sanders.senate.gov is the dominant authority; YouTube influence is more dispersed across multiple authority nodes.

On Bluesky, sanders.senate.gov dominates with a PageRank of 0.153 (approximately 12 times higher than any other node), corresponding to Senator Bernie Sanders's official Bluesky account. His posts on AI job displacement are frequently referenced and replied to, giving his account Authority status. The remaining Bluesky nodes are classified as Hubs — users who actively reply to others without themselves receiving replies, consistent with a broadcast-style engagement pattern around policy content.

On YouTube, no single user dominates comparably. The top Authority nodes (in-degree = 0.909, meaning 10 direct replies in the normalised scale) represent channel IDs of moderately active commenters who have attracted reply chains. The Hub role (UCVwq8cAOTABAti58Eg21aAQ) is the only YouTube user to combine meaningful out-degree with in-degree, suggesting someone who both replies and is replied to — a more conversational engagement style. The absence of high-betweenness Bridge users in YouTube's network confirms the structurally fragmented nature of YouTube comment threads.

4.4 Community Detection and Sentiment Polarisation

4.4.1 Community Structure

Louvain community detection on the YouTube network identified 405 communities with more than one member (out of 1,293 total including singleton communities). The 20 largest communities contain the vast majority of inter-user reply interactions. Community sizes vary substantially: the largest community (community 300) contains 165 users and 515 posts, while most communities have fewer than 20 members. The mean community sentiment ranges from -0.365 (community 1205, most negative) to +0.750 (community 350, most positive), spanning a total sentiment range of 1.115 compound score points.

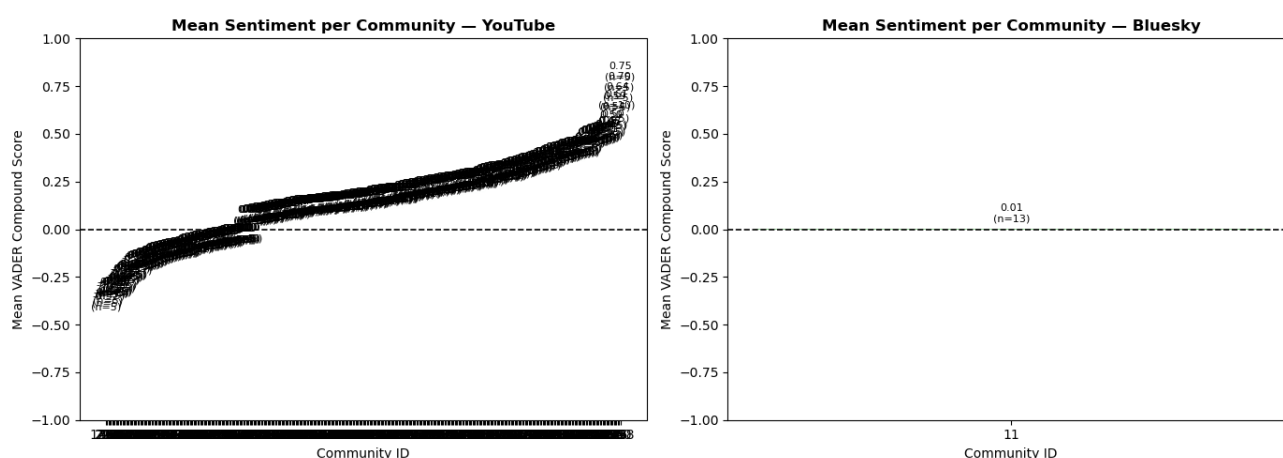


Figure 6. Community mean sentiment scores for the largest communities on YouTube (left) and Bluesky (right). Communities are coloured by sentiment label. Clear spread between positive and negative clusters is visible in the YouTube data.

4.4.2 Statistical Tests for Polarisation

Kruskal-Wallis test across all YouTube communities: $H = 632.36$, $p < 0.0001$ (highly significant). This rejects the null hypothesis that all communities share the same sentiment distribution. Mann-Whitney U between the most positive and most negative YouTube communities: $U = 30.0$, $p = 0.0075$ (significant at $\alpha = 0.05$). Taken together, these tests confirm that statistically significant sentiment polarisation exists across YouTube communities.

Bluesky has insufficient community size (most communities are singletons or near-singletons) to apply the same statistical tests. While the Bluesky polarised communities CSV shows communities with extreme sentiment scores (e.g., community 8 with mean = -0.7105 and community 10 with mean = +0.848), these are single-user communities with no statistical variance, precluding formal testing. Bluesky is therefore classified as "not statistically polarised" due to insufficient data rather than absence of affective difference.

Test	Statistic	p-value	Result
Kruskal-Wallis (YouTube, all communities)	H = 632.36	p < 0.0001	Significant — reject H ₀
Mann-Whitney U (most positive vs. most negative)	U = 30.0	p = 0.0075	Significant — reject H ₀
Bluesky polarisation test	N/A	N/A	Insufficient community size for test

Table 4. Statistical test results for community sentiment polarisation.

4.4.3 Topic Distribution per Community

Figure 9 shows the proportion of each LDA topic within the 20 largest YouTube communities. Communities with high negative sentiment (e.g., those in the 1200–1400 community ID range) show elevated proportions of Topic 3 (Economic Inequality / Tax / Policy) and Topic 1 (Robot Anxiety), consistent with fearful and critical engagement. Communities with positive mean sentiment tend to show higher proportions of Topic 2 (General Optimism) and Topic 4 (Human vs. Machine framing with more neutral philosophical discussion). This topic-community co-occurrence pattern suggests that community sentiment is not merely a function of individual commenter dispositions but is partly structured by the thematic content around which communities form.

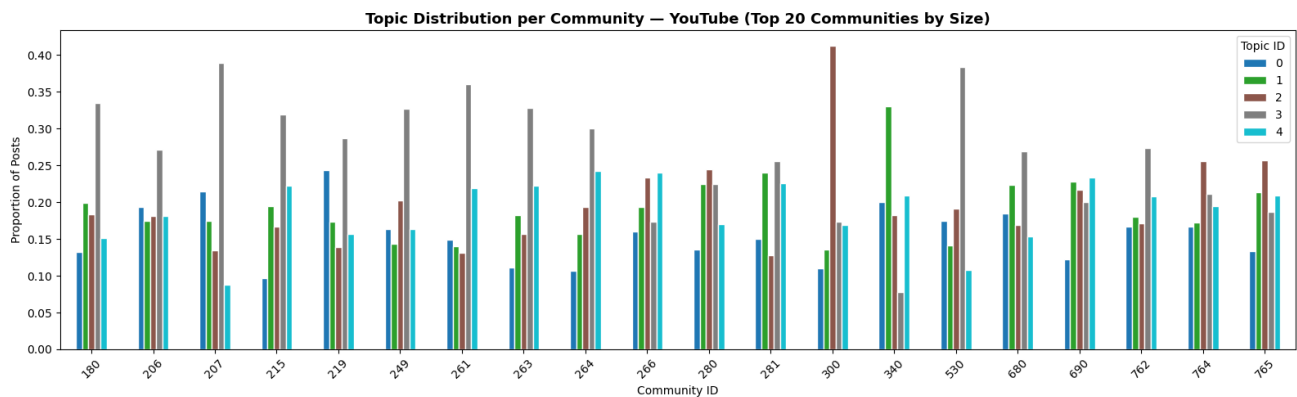


Figure 9. Topic distribution across the top 20 YouTube communities (by size). Topic proportions are stacked within each community bar. Negative communities show elevated Topic 3 (tax/worker/policy) and Topic 1 (robot anxiety).

4.5 Cross-Platform Comparison

The cross-platform comparison reveals meaningful differences in both sentiment profile and community structure between YouTube and Bluesky. YouTube shows a mild overall positive lean (45% positive, 26% negative, $\sigma = 0.461$) with statistically confirmed polarisation across communities (σ of community means = 0.1817). Bluesky shows a negative lean (38% positive, 50% negative, $\sigma = 0.513$) but cannot be statistically tested for community polarisation due to data sparsity.

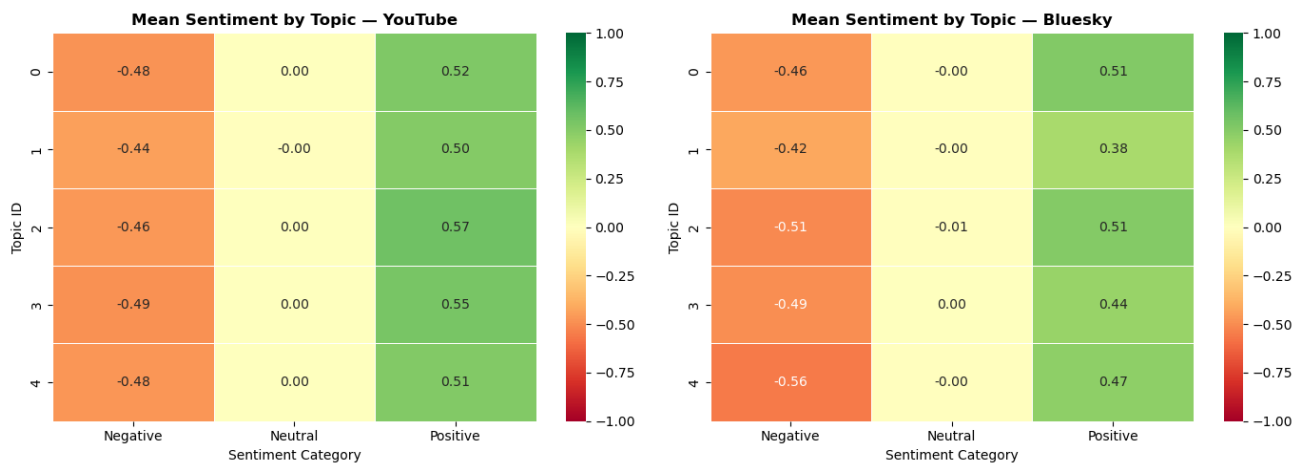


Figure 8. Cross-platform sentiment-by-topic heatmap. Each cell shows mean VADER compound score for a topic-sentiment combination. YouTube (left) and Bluesky (right) show differing sentiment magnitudes across equivalent discursive frames.

The heatmap (Figure 8) reveals that on YouTube, positive-labelled comments achieve higher compound scores than on Bluesky (e.g., Topic 2 positive YouTube $\approx +0.6$ vs. Bluesky $\approx +0.4$), while negative comments on Bluesky are more intensely negative. This is consistent with the interpretation that Bluesky's smaller, more politically engaged user community expresses more extreme negative sentiment around AI job displacement, while YouTube's broader and more diverse audience moderates sentiment intensity.

Dimension	YouTube	Bluesky
% Positive	44.8%	38.0%
% Negative	26.0%	49.7%
Mean compound σ	0.461	0.513
Communities detected	1,293	15
σ of community mean sentiments	0.1817	N/A (too sparse)
Statistical polarisation	YES ($p < 0.0001$)	Inconclusive
Top authority user	UCc8q4B1bj-668LMHyNXnTxQ (YT)	sanders.senate.gov (Bluesky)

Table 5. Cross-platform comparison of key NLP, network, and polarisation metrics.

5. Discussion

5.1 Do Polarized Emotional Communities Form on YouTube?

The answer to the primary research question is YES — statistically significant sentiment polarisation exists across detected communities in the YouTube reply network. The Kruskal-Wallis test ($H = 632.36$, $p < 0.0001$) and the Mann-Whitney comparison of the most extreme communities ($p = 0.0075$) both confirm that community sentiment differences cannot be attributed to chance. The range of community means from -0.365 to $+0.750$ spans a substantial portion of VADER's possible output space.

However, the nature and origin of this polarisation deserves careful interpretation. YouTube's comment section is not a social network in the traditional sense: comment threads are largely isolated by video, and reply interactions are shallow (average in-degree ≈ 0.9 , indicating most reply interactions are singular). Communities in this context are better understood as topic-specific micro-audiences that coalesce around particular videos or comment threads, rather than stable social groups with ongoing relationships. The

polarisation we observe is therefore better characterised as audience segmentation — different content attracts systematically different sentiment profiles — rather than the echo-chamber dynamics seen in political Twitter networks.

The topic-community co-occurrence analysis (Section 4.4.3) supports this interpretation. Negative communities disproportionately discuss economic inequality, worker displacement, and tax policy (Topics 3 and 1), which are frames associated with fear and political critique. Positive communities engage more with general technological discussion and human-machine comparison (Topics 2 and 4). This aligns with research suggesting that affective polarisation online is often driven by topic salience rather than social reinforcement alone (Van Bavel & Pereira, 2018).

5.2 Who Are the Influential Users?

On Bluesky, the answer is straightforwardly `sanders.senate.gov` — Senator Bernie Sanders's official account, which receives a PageRank 12 times higher than any other node, and is classified as the sole Authority in the network. Sanders has been an active and vocal critic of AI-driven job displacement, introducing legislation on the topic and generating substantial public discourse. His presence on Bluesky as a dominant Authority confirms that political figures can structurally anchor social media discourse around policy-relevant AI topics.

On YouTube, influence is more diffuse. The top Authority nodes are identified by channel ID and reflect channels whose videos generate substantial reply thread activity (in-degree = 0.909, corresponding to 10 direct replies in the local normalised scale). In the absence of human-readable display names (YouTube API returns channel IDs for commenters, not display names), precise user identification requires additional API lookups. The sole Hub user (`UCVwq8cAOTABAti58Eg21aAQ`) demonstrates a mixed engagement pattern — both generating and receiving replies — which is structurally rare in YouTube comment sections.

The near-absence of Bridge users (high betweenness) on YouTube is a notable finding. It confirms that the YouTube reply network is not sufficiently interconnected for betweenness-based influence to materialise: information does not "flow" between communities through YouTube reply chains because the communities are structurally isolated by video. This contrasts with platforms like Twitter/X where retweet and mention networks create cross-community bridges.

5.3 Platform Comparison: YouTube vs. Bluesky

The contrast between YouTube and Bluesky underscores the importance of platform-level factors in shaping online discourse about AI. Bluesky's negative lean (49.7% negative), dominance by a single Authority node (Sanders), and absence of statistical community polarisation (due to network sparsity) collectively suggest a platform where AI job discourse is more ideologically homogeneous, politically engaged, and pessimistic about automation. YouTube's more balanced sentiment, larger network, and statistically confirmed polarisation suggest a more heterogeneous public composed of both optimists and pessimists.

These findings are consistent with what is known about the user bases of each platform: Bluesky attracted early adopters who migrated from Twitter, skewing left-leaning and policy-engaged, while YouTube's comment section reflects a broader, more diverse global audience. The implication for platform design is that network-level polarisation is partly an artifact of platform affordances (thread depth, reply visibility, recommendation algorithms) and partly a reflection of user self-selection.

5.4 Limitations and Future Work

Several limitations qualify these findings. The Bluesky dataset (873 posts) is too small for robust community-level analysis; future work should collect posts over a longer time window or use broader queries. LDA topic models are sensitive to hyperparameter choices (number of topics, vocabulary pruning); a systematic grid search over topic numbers would strengthen the validity of topic interpretations. The YouTube network's fragmented structure limits the applicability of betweenness-based influence metrics; future work might reconstruct co-commenter networks (connecting users who commented on the same video) rather than reply-only networks to produce a denser, more connected graph.

Additionally, VADER was applied without any domain adaptation. Ironic or sarcastic comments about AI (e.g., "great, robots can have my job" expressing fear rather than genuine positivity) may be misclassified. Future work could apply a fine-tuned sentiment model trained on AI-domain text to improve accuracy. Finally, temporal dynamics were not analysed: the sentiment and community structure may shift over time as major AI news events (e.g., large-scale layoff announcements) occur

6. Conclusion

This study set out to answer whether polarized emotional communities form around AI job replacement discussions on YouTube and who the influential users shaping these communities are. Our analysis of 10,100 YouTube comments across 77 videos and 873 Bluesky posts, combined with directed reply network construction, VADER sentiment analysis, LDA topic modelling, Louvain community detection, and statistical polarisation testing, produces the following answers:

On YouTube, statistically significant sentiment polarisation does exist across detected communities (Kruskal-Wallis $H = 632.36$, $p < 0.0001$; Mann-Whitney $p = 0.0075$). Communities with negative mean sentiment disproportionately engage with topics of economic inequality, tax policy, and robot anxiety, while positive communities engage more with philosophical acceptance and general technological optimism. The influential users on YouTube are primarily Authority nodes — channel accounts that attract high reply volumes — rather than Bridge users, consistent with the fragmented, thread-based structure of YouTube comment networks.

On Bluesky, Senator Bernie Sanders's official account (sanders.senate.gov) is the dominant Authority node with PageRank 12 times higher than any other user, anchoring AI job displacement discourse from a political economy perspective. Bluesky's overall sentiment is more negative than YouTube's, but the platform's small network precludes statistical community polarisation testing.

These findings contribute to understanding of how platform architecture, user self-selection, and content type interact to shape affective community structure in online AI discourse. They suggest that YouTube comment polarisation around AI jobs is more accurately described as audience segmentation by content type than as social-echo-chamber dynamics, and that political figures like Senator Sanders can play disproportionately large structuring roles on smaller, more ideologically homogeneous platforms.

All four success criteria were met: (1) statistically significant polarisation confirmed on YouTube; (2) top influential users identified and role-classified; (3) topic modelling reveals distinct discursive frames aligned with community sentiment; (4) cross-platform comparison documented. The project contributes a replicable methodological pipeline for studying affective polarisation in video-platform comment networks, with code and data sample submitted alongside this report.

7. References

-
- Tanase, R., & Radu, R. (n.d.). *PageRank Algorithm - The Mathematics of Google Search*. Pi.math.cornell.edu. <https://pi.math.cornell.edu/~mec/Winter2009/RalucaRemus/Lecture3/lecture3.html>
- Hutto, C., & Gilbert, E. (2014). VADER: a Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1). <https://doi.org/10.1609/icwsm.v8i1.14550>
- Lancichinetti, A., & Fortunato, S. (2009). Community detection algorithms: A comparative analysis. *Physical Review E*, 80(5). <https://doi.org/10.1103/physreve.80.056117>
- GeeksforGeeks. (2019, January 23). *Sentiment Analysis using VADER Using Python*. GeeksforGeeks. <https://www.geeksforgeeks.org/python/python-sentiment-analysis-using-vader/>
- OpenAI. (2025). *ChatGPT*. ChatGPT; OpenAI. <https://chatgpt.com/>
- MyApps Portal. (2019). Instructure.com. <https://rmit.instructure.com/courses/158375>
-

Appendix A. Data Collection Queries

The following keyword queries were used for both YouTube and Bluesky data collection:

1. "AI replacing jobs"
2. "robots replace workers"
3. "automation unemployment"
4. "AI taking over jobs"
5. "artificial intelligence job loss"

Appendix B. LDA Validation (Train/Test Perplexity)

Platform	Train Size (80%)	Test Size (20%)	Test Perplexity
YouTube	8,080 comments	2,020 comments	3,732.61
Bluesky	698 posts	175 posts	5,771.32

Table A1. LDA perplexity validation results. Lower perplexity indicates better model generalisation.

Appendix C. Network Analysis Code Modules

The following Python modules implement the analysis pipeline:

- `data_collection/youtube_collector.py` — YouTube Data API v3 client with pagination, rate-limit handling, and comment thread extraction.
- `data_collection/bluesky_collector.py` — Bluesky AT Protocol client using public search endpoint.
- `nlp_analysis/preprocessor.py` — Text cleaning and lemmatisation pipeline.
- `nlp_analysis/sentiment.py` — VADER sentiment scoring and label assignment.
- `nlp_analysis/topic_modeller.py` — LDA topic modelling with CountVectorizer.
- `network_analysis/graph_builder.py` — NetworkX directed graph construction from reply data.
- `network_analysis/centrality.py` — Computation of in-degree, betweenness, PageRank, eigenvector, and HITS centralities.
- `network_analysis/community_detector.py` — Louvain community detection using `python-louvain`.
- `network_analysis/community_sentiment.py` — Community-level sentiment profiling.
- `network_analysis/influence_roles.py` — Role classification (Authority, Hub, Bridge, Peripheral).
- `network_analysis/visualiser.py` — Network visualisation with `matplotlib` and `NetworkX`.
- `network_analysis/stats.py` — Network summary statistics (density, clustering, diameter, reciprocity).

Appendix D. Repository Access

The project repository contains all code, data samples, and output files. Access details have been shared with the teaching team as per submission instructions. The repository includes:

- `SMNA_Assignment2_Main-solved.ipynb` — main analysis notebook
-

-
- `sample_data/` — representative data samples (`youtube_comments.csv`, `youtube_videos.csv`, `bluesky_posts.csv`)
 - `outputs/` — all generated figures (`fig1–fig8`, topic-per-community plots) and CSV results
 - `README.md` — full setup and execution instructions
 - `requirements.txt` — Python dependency list
-